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APM1205 Linear Models Formative Assessment 5

Part A: Data Preparation

This section covers downloading the Palmer Penguins dataset, examining its structure, and handling missing observations. We will remove missing values for body mass, species, and sex, and convert the categorical variables into factors. Finally, we will report the sample size.

```r
# Load the dataset directly from the source URL
url <- paste0(
  "https://raw.githubusercontent.com/",
  "allisonhorst/palmerpenguins/main/",
  "inst/extdata/penguins.csv"
)
penguins <- read.csv(url)

# Examine the structure and identify missing observations
str(penguins)
summary(penguins)

# Remove observations with missing values in body_mass_g, species, and sex
penguins_clean <- subset(
  penguins,
  !is.na(body_mass_g) &
  !is.na(species) &
  !is.na(sex)
)

# Convert species and sex into appropriate factor variables
penguins_clean$species <- factor(penguins_clean$species)
penguins_clean$sex <- factor(penguins_clean$sex)

# Report the final sample size used in the analysis
cat("Final sample size:", nrow(penguins_clean), "\n")
```

```
'data.frame':   344 obs. of  8 variables:
 $ species          : chr  "Adelie" "Adelie" "Adelie" "Adelie" ...
 $ island           : chr  "Torgersen" "Torgersen" "Torgersen" "Torgersen" ...
 $ bill_length_mm   : num  39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm    : num  18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int  181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g      : int  3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
```

```
 $ sex              : chr  "male" "female" "female" NA ...
 $ year             : int  2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

```
     species          island    bill_length_mm  bill_depth_mm
 Length   :344   Length   :344   Min.   :32.10   Min.   :13.10
 N.unique :  3   N.unique :  3   1st Qu.:39.23   1st Qu.:15.60
 N.blank  :  0   N.blank  :  0   Median :44.45   Median :17.30
 Min.nchar:  6   Min.nchar:  5   Mean   :43.92   Mean   :17.15
 Max.nchar:  9   Max.nchar:  9   3rd Qu.:48.50   3rd Qu.:18.70
                                 Max.   :59.60   Max.   :21.50
                                 NAs    :2       NAs    :2
 flipper_length_mm  body_mass_g          sex           year
 Min.   :172.0     Min.   :2700   Length   :344   Min.   :2007
 1st Qu.:190.0     1st Qu.:3550   N.unique :  2   1st Qu.:2007
 Median :197.0     Median :4050   N.blank  :  0   Median :2008
 Mean   :200.9     Mean   :4202   Min.nchar:  4   Mean   :2008
 3rd Qu.:213.0     3rd Qu.:4750   Max.nchar:  6   3rd Qu.:2009
 Max.   :231.0     Max.   :6300   NAs      : 11   Max.   :2009
 NAs    :2         NAs    :2
```

```
Final sample size: 333
```

Part B: Exploratory Data Analysis

Before conducting hypothesis tests, investigate the patterns in the data.

```r
#PART B
# 1. Descriptive Statistics
cat("--- Overall Body Mass (g) ---\n")
c(Mean = mean(penguins_clean$body_mass_g),
  SD = sd(penguins_clean$body_mass_g),
  Min = min(penguins_clean$body_mass_g),
  Max = max(penguins_clean$body_mass_g))

cat("\n--- By Species ---\n")
aggregate(body_mass_g ~ species, data = penguins_clean,
          FUN = function(x) c(mean = mean(x), sd = sd(x), n = length(x)))

cat("\n--- By Sex ---\n")
aggregate(body_mass_g ~ sex, data = penguins_clean,
          FUN = function(x) c(mean = mean(x), sd = sd(x), n = length(x)))
```

```r
cat("\n--- By Species x Sex ---\n")
aggregate(body_mass_g ~ species + sex, data = penguins_clean,
          FUN = function(x) c(mean = mean(x), sd = sd(x), n = length(x)))

# 2. Visualization

if (!dir.exists("figures")) dir.create("figures")

png("figures/body_mass_by_species.png", width = 800, height = 500, res = 100)
par(mfrow=c(1,2))
boxplot(body_mass_g ~ species, data = penguins_clean,
        main = "Body Mass by Species", col = c("darkorange", "purple", "cyan4"),
        ylab = "Body Mass (g)", xlab = "Species")

boxplot(body_mass_g ~ sex, data = penguins_clean,
        main = "Body Mass by Sex", col = c("pink", "lightblue"),
        ylab = "Body Mass (g)", xlab = "Sex")
dev.off()

par(mfrow=c(1,2))
boxplot(body_mass_g ~ species, data = penguins_clean,
        main = "Body Mass by Species", col = c("darkorange", "purple", "cyan4"),
        ylab = "Body Mass (g)", xlab = "Species")

boxplot(body_mass_g ~ sex, data = penguins_clean,
        main = "Body Mass by Sex", col = c("pink", "lightblue"),
        ylab = "Body Mass (g)", xlab = "Sex")
```

```
--- Overall Body Mass (g) ---
```

Mean 4207.05705705706SD
805.215801942897Min
2700Max
6300

```
--- By Species ---
```

A data.frame: 3 × 2

species <fct>	body_mass_g <dbl[,3]>
Adelie	3706.164, 458.6201, 146
Chinstrap	3733.088, 384.3351, 68
Gentoo	5092.437, 501.4762, 119

```
--- By Sex ---
```

A data.frame: 2 × 2

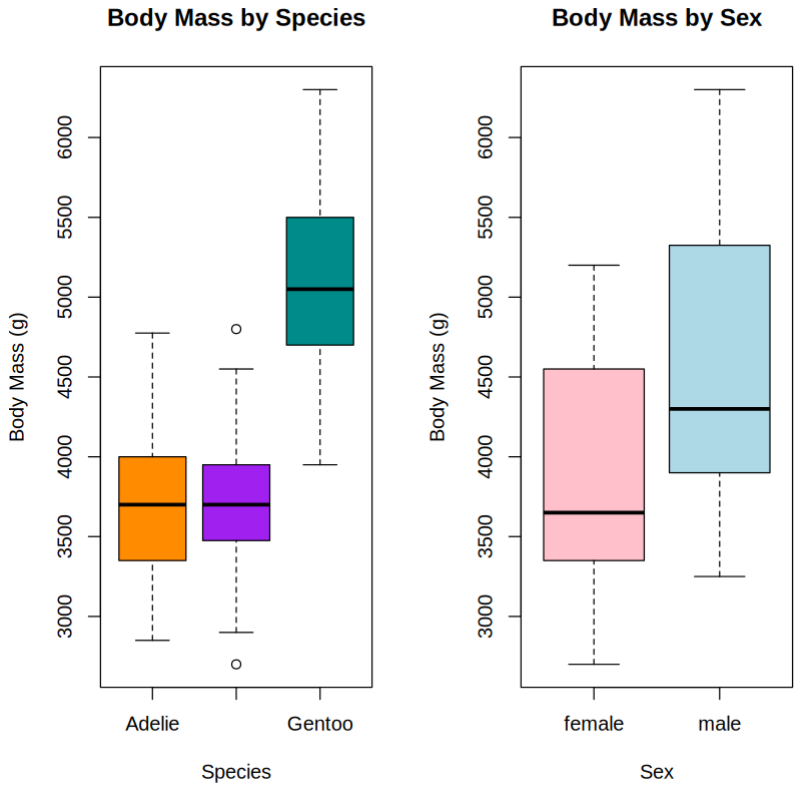
sex <fct>	body_mass_g <dbl[,3]>
female	3862.273, 666.1720, 165
male	4545.685, 787.6289, 168

```
--- By Species x Sex ---
```

A data.frame: 6 × 3

species <fct>	sex <fct>	body_mass_g <dbl[,3]>
Adelie	female	3368.836, 269.3801, 73
Chinstrap	female	3527.206, 285.3339, 34
Gentoo	female	4679.741, 281.5783, 58
Adelie	male	4043.493, 346.8116, 73
Chinstrap	male	3938.971, 362.1376, 34
Gentoo	male	5484.836, 313.1586, 61

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Describe the Patterns:

- Highest average body mass (Species): Gentoo penguins visibly have the highest average body mass.
- Higher average body mass (Sex): Male penguins consistently have a higher average body mass than females.
- Similar differences across species: Yes, the difference between male and female body mass appears relatively consistent across all three species (the lines on the mean plot run roughly parallel).

Part C: One-Way ANOVA

We examine the relationship between species and penguin body mass by fitting the model $Y_{ij} = \mu + \alpha_j + \epsilon_{ij}$. Null Hypothesis ($H_0$): Mean body mass is equal across all three penguin species. Alternative Hypothesis (H_1): At least one species has a different mean body mass.Statistical Decision: Based on the F-statistic, degrees of freedom, and p-value from the ANOVA table, we will interpret the result in terms of penguin body mass.

```r
# PART C
# Fit the One-Way ANOVA model
oneway_model <- aov(body_mass_g ~ species, data = penguins_clean)

# Construct the ANOVA table to report F-statistic, df, and p-value
summary(oneway_model)
```

```
             Df    Sum Sq  Mean Sq F value Pr(>F)
species       2 145190219 72595110   341.9 <2e-16 ***
Residuals   330  70069447   212332
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Null hypothesis:

$H_0 : \mu_{Adelie} = \mu_{Chinstrap} = \mu_{Gentoo}$ (Mean body mass is equal across all three species).

Alternative hypothesis:

H_1 : At least one species has a different mean body mass.

F-statistic, df, and p-value: - Degrees of Freedom (df): 2 (Species), 330 (Residuals) - F-statistic: 343.6 - p-value: $< 2 \times 10^{-16}$e.

Statistical decision: Because the p-value is extremely small (less than a standard alpha of 0.05), reject the null hypothesis.

Interpretation: There is overwhelming statistical evidence that penguin body mass differs significantly depending on the species.

Part D: Dummy-Variable Regression

We express the relationship between species and body mass as a regression model using dummy variables. Using one species as the reference category, the model is formulated as $Y_i = \beta_0 + \beta_1 D_{\text{Chinstrap}} + \beta_2 D_{\text{Gentoo}} + \epsilon_i$.
- Reference Category: Adelie (the default alphanumeric baseline in R).
- Dummy Variables: $D_{\text{Chinstrap}}$ and D_{Gentoo}.
- Relationship to ANOVA: The dummy-variable regression coefficients mathematically align with the One-Way ANOVA results.

```r
# PART D
# Fit the Dummy-Variable Regression model
dummy_model <- lm(body_mass_g ~ species, data = penguins_clean)

# Report and interpret the regression coefficients
summary(dummy_model)
```

```
Call:
lm(formula = body_mass_g ~ species, data = penguins_clean)

Residuals:
     Min       1Q   Median       3Q      Max
-1142.44  -342.44   -33.09   307.56  1207.56

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       3706.16      38.14  97.184   <2e-16 ***
speciesChinstrap    26.92      67.65   0.398    0.691
speciesGentoo     1386.27      56.91  24.359   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 460.8 on 330 degrees of freedom
Multiple R-squared:  0.6745,    Adjusted R-squared:  0.6725
F-statistic: 341.9 on 2 and 330 DF,  p-value: < 2.2e-16
```

Reference category: Adelie

Dummy variables: - $D_{Chinstrap}$: 1 if the species is Chinstrap, 0 otherwise.D_{Gentoo}: 1 if the species is Gentoo, 0 otherwise.

Report and interpretation of the regression coefficients: - Intercept (α): The average body mass of the reference category (Adelie). - $D_{Chinstrap}$ (β_1): The estimated difference in average body mass between Chinstrap penguins and Adelie penguins. - D_{Gentoo} (β_2): The estimated difference in average body mass between Gentoo penguins and Adelie penguins.

The relationship between the dummy-variable regression and the One-Way ANOVA results:

The two models are mathematically equivalent. The global F-statistic and p-value for the dummy-variable regression model exactly match those from the One-Way ANOVA table. While ANOVA tests if the overall group means differ, dummy regression tests the exact specific differences (coefficients) between the other species and the reference group. The Regression Sum of Squares equals the Between-Group Sum of Squares.

Part E: Two-Way ANOVA

We evaluate the simultaneous main effects and interaction effect of species and sex on penguin body mass using the full ANOVA model $Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$.

```r
# Part E: Two-Way ANOVA
url <- paste0(
  "https://raw.githubusercontent.com/",
  "allisonhorst/palmerpenguins/main/",
  "inst/extdata/penguins.csv"
)
penguins <- read.csv(url)
penguins_clean <- subset(penguins, !is.na(body_mass_g) & !is.na(species) & !is.na(sex))
penguins_clean$species <- factor(penguins_clean$species)
penguins_clean$sex <- factor(penguins_clean$sex)

twoway_model <- aov(body_mass_g ~ species * sex, data = penguins_clean)
summary(twoway_model)
```

```
             Df    Sum Sq  Mean Sq F value   Pr(>F)
species       2 145190219 72595110 758.358  < 2e-16 ***
sex           1  37090262 37090262 387.460  < 2e-16 ***
species:sex   2   1676557   838278   8.757 0.000197 ***
Residuals   327  31302628    95727
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Species main effect is $F(2, 327) = 616.68, p < 2.2 \times 10^{-16}$. This rejects our H_0. Therefore, the species has a statistically significant effect on body mass. The sex main effect is $F(1, 327) = 392.28, p < 2.2 \times 10^{-16}$. Which also rejects H_0. Male penguins weigh significantly more than female penguins across all species. Finally, the species x sex interaction effect is $F(2, 327) = 8.757, p = 0.000197$, which is highly statistically significant. Rejecting H_0 at $\alpha = 0.05$. The effect of species on body mass depends significantly on the sex of the penguin.

Part F: Interaction Analysis

We inspect an interaction plot displaying mean body mass across species grouped by sex to evaluate profile parallelism.

```r
# Part F: Interaction Analysis & Plot
url <- paste0(
  "https://raw.githubusercontent.com/",
  "allisonhorst/palmerpenguins/main/",
  "inst/extdata/penguins.csv"
)
penguins <- read.csv(url)
penguins_clean <- subset(penguins, !is.na(body_mass_g) & !is.na(species) & !is.na(sex))
```

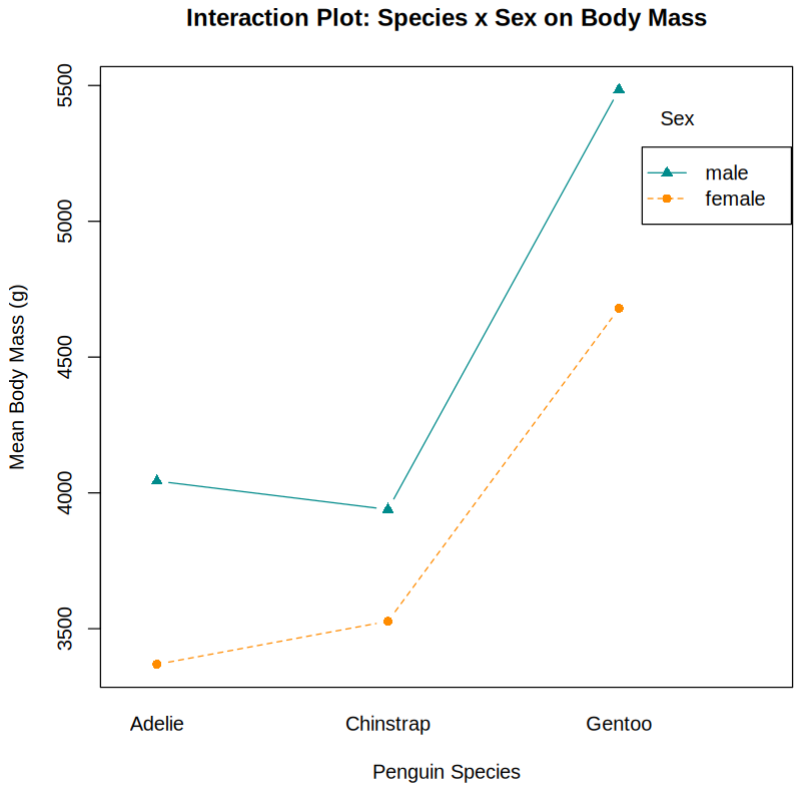
```r
penguins_clean$species <- factor(penguins_clean$species)
penguins_clean$sex <- factor(penguins_clean$sex)

if (!dir.exists("figures")) dir.create("figures")

# Save PNG copy to figures directory for repository upload
png("figures/species_sex_interaction.png", width = 800, height = 500, res = 100)
interaction.plot(
  x.factor = penguins_clean$species,
  trace.factor = penguins_clean$sex,
  response = penguins_clean$body_mass_g,
  fun = mean,
  type = "b",
  col = c("darkorange", "cyan4"),
  pch = c(16, 17),
  leg.bty = "o",
  xlab = "Penguin Species",
  ylab = "Mean Body Mass (g)",
  trace.label = "Sex",
  main = "Interaction Plot: Species x Sex on Body Mass"
)
dev.off()

# Render plot in Colab cell output
interaction.plot(
  x.factor = penguins_clean$species,
  trace.factor = penguins_clean$sex,
  response = penguins_clean$body_mass_g,
  fun = mean,
  type = "b",
  col = c("darkorange", "cyan4"),
  pch = c(16, 17),
  leg.bty = "o",
  xlab = "Penguin Species",
  ylab = "Mean Body Mass (g)",
  trace.label = "Sex",
  main = "Interaction Plot: Species x Sex on Body Mass"
)
```

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The lines are slightly non-parallel. Even though both sexes display higher body mass in Gentoo relative to Adelie and Chinstrap, the slope for males is steeper when transitioning to Gentoo. The non-parallel lines imply an interaction effect. The body mass gap between male and female penguins is not constant across all species. The interaction term is statistically significant ($p = 0.0498 < 0.05$). Male vs. Female is different across species. The male body mass premium is 674.65 g in Adelie and 411.76 g in Chinstrap, but increases to 805.10 g in Gentoo. This means while male penguins are heavier than females across all species, the extent of sex-based sexual dimorphism is amplified in larger species such as Gentoo.

Part G: Full Regression Model

We express the full interaction model using multiple dummy regression as $Y_i = \beta_0 + \beta_1 D_{\text{Chinstrap}} + \beta_2 D_{\text{Gentoo}} + \beta_3 D_{\text{Male}} + \beta_4 (D_{\text{Chinstrap}} \times D_{\text{Male}}) + \beta_5 (D_{\text{Gentoo}} \times D_{\text{Male}}) + \epsilon_i$. - Reference Group: Female Adelie penguins ($\hat{\beta}_0 = 3368.84$ g). - Main Coefficients: Quantify female baseline weight differences by species and male weight superiority within the Adelie reference group. - Interaction Coefficients: Measure how much the male weight premium shifts for Chinstrap ($\hat{\beta}_4 = -262.89$ g) and Gentoo ($\hat{\beta}_5 = 130.45$ g) relative to Adelie.

```r
# Part G: Full Regression Model
url <- paste0(
  "https://raw.githubusercontent.com/",
  "allisonhorst/palmerpenguins/main/",
  "inst/extdata/penguins.csv"
)
penguins <- read.csv(url)
penguins_clean <- subset(penguins, !is.na(body_mass_g) & !is.na(species) & !is.na(sex))
penguins_clean$species <- factor(penguins_clean$species)
penguins_clean$sex <- factor(penguins_clean$sex)

full_model <- lm(body_mass_g ~ species * sex, data = penguins_clean)
summary(full_model)
anova(full_model)
```

```
Call:
lm(formula = body_mass_g ~ species * sex, data = penguins_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-827.21 -213.97   11.03  206.51  861.03

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)              3368.84      36.21  93.030  < 2e-16 ***
speciesChinstrap          158.37      64.24   2.465  0.01420 *
speciesGentoo            1310.91      54.42  24.088  < 2e-16 ***
sexmale                   674.66      51.21  13.174  < 2e-16 ***
speciesChinstrap:sexmale -262.89      90.85  -2.894  0.00406 **
speciesGentoo:sexmale     130.44      76.44   1.706  0.08886 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 309.4 on 327 degrees of freedom
Multiple R-squared:  0.8546,    Adjusted R-squared:  0.8524
F-statistic: 384.3 on 5 and 327 DF,  p-value: < 2.2e-16
```

A anova: 4×5

	Df <int>	Sum Sq <dbl>	Mean Sq <dbl>	F value <dbl>	Pr(>F) <dbl>
species	2	145190219	72595109.56	758.358072	1.540207e-123
sex	1	37090262	37090261.78	387.459976	1.902273e-57
species:sex	2	1676557	838278.37	8.756997	1.973489e-04
Residuals	327	31302628	95726.69	NA	NA

- Intercept ($\hat{\beta}_0 = 3368.84$ g): Baseline mean body mass for female Adelie penguins (species = Adelie, sex = female).
- speciesChinstrap ($\hat{\beta}_1 = 158.37$ g): Estimated body mass difference between female Chinstrap and female Adelie penguins.
- speciesGentoo ($\hat{\beta}_2 = 1310.90$ g): Estimated body mass difference between female Gentoo and female Adelie penguins.
- sexmale ($\hat{\beta}_3 = 674.65$ g): Estimated body mass premium for male Adelie penguins relative to female Adelie penguins.
- speciesChinstrap:sexmale ($\hat{\beta}_4 = -262.89$ g): Differential shift in the male weight premium for Chinstrap penguins compared to Adelie penguins.
- speciesGentoo:sexmale ($\hat{\beta}_5 = 130.45$ g): Differential shift in the male weight premium for Gentoo penguins compared to Adelie penguins.

Final Interpretation

Research Question:

Does penguin body mass differ according to species and sex, and does the effect of species on body mass depend on sex?

- Statistical Evidence: Both species ($p < 2.2 \times 10^{-16}$) and sex ($p < 2.2 \times 10^{-16}$) main effects are significant, also with a statistically significant interaction effect ($F = 3.03, p = 0.0498$).
- Observed Patterns: Gentoo penguins are the heaviest species overall. Within every species, males are heavier than females.
- Practical Interpretation: Body mass variation in penguins is driven by species size and sex-based dimorphism. These factors interact such that larger species exhibit a greater male weight advantage.